

Functions : A function is a reusable code that performs specific tasks

Types of functions :

- Built-in functions : Already available in Python
- User-defined functions : Created by users
- Lambda (Anonymous) : Defined without a name

```
In [2]: def students_name(name):  
        print("Students name is ", name)  
  
        students_name("yasir")
```

Students name is yasir

```
In [3]: # Write a function to print "Hello, World!".  
  
def name():  
    print("hello world")  
  
name()
```

hello world

```
In [4]: # Write a function that takes a name as input and prints "Hello, <name>".  
  
def x(name):  
    print("hello " + name)  
  
x("yasir")
```

hello yasir

```
In [5]: # Write a function that adds two numbers and returns the result.  
  
def add(x, y):  
    return x+y  
  
add(4, 4)
```

Out[5]: 8

```
In [6]: # Write a function that checks whether a number is even or odd.  
  
def check(num):  
    if num % 2 == 0:  
        print(f'{num} is even')  
    else:  
        print(f'{num} is odd')  
check(4)  
check(7)
```

4 is even

7 is odd

In [7]: *# Write a function that calculates the square of a number.*

```
def calculate(num):  
    return num*num  
  
print(calculate(2))  
print(calculate(25))
```

4
625

In [8]: *# Write a function that finds the maximum of two numbers.*

```
def num(a,b):  
    if a>b:  
        print(f'{a} is greater')  
    else:  
        print(f'{b} is greater')  
  
num(2,3)  
num(32,532)
```

3 is greater
532 is greater

In [9]: **def** add(a,b):
 return a+b

```
x = add(3,2)  
  
print(x)
```

5

In [10]: **def** x(name="yasir"):
 print("hello" , name)

```
x()
```

hello yasir

In [11]: **def** students(name,age):
 print(name,age)

students(name="yasir",age=20)

yasir 20

In [12]: **def** add_all(*args):
 return sum(args)

add_all(1,2,3)

Out[12]: 6

In [13]: **def** display(**kwargs):
 for key , value **in** kwargs.items():
 print(key , value)

```
display(name="yasir",age=20)
```

```
name yasir  
age 20
```

```
In [14]: x = 10  #Global variable  
  
def func():  
  
    x = 5  #local variable  
    print(x)  
  
func()  
  
print(x)
```

```
5  
10
```

```
In [15]: x = 5  
  
def x():  
    global x  
    x = 10  
  
x()  
print(x)
```

```
10
```

```
In [16]: def x():  
    def y():  
        print("nested loop")  
    y()  
x()
```

```
nested loop
```

```
In [17]: def factorial(n):  
    if n== 1 or n==0:  
        return 1  
    return n * factorial(n-1)  
n = 5  
print(factorial(n))
```

```
120
```

```
In [18]: squ = lambda x: x*x  
print(squ(5))
```

```
25
```

```
In [19]: add = lambda x,y : x+y  
print(add(5,3))
```

```
8
```

```
In [20]: def apply(x,y):  
         return x(y)  
         print(apply(lambda z:z**2,45))
```

2025

```
In [21]: num = [1,2,3]  
         print(list(map(lambda x: x*x , num)))
```

[1, 4, 9]

```
In [22]: c = list(filter(lambda x : x%2==0 , num))  
         print(c)
```

[2]

```
In [24]: from functools import reduce  
         total = reduce(lambda x, y: x + y, num)  
         print(total)
```

6

```
In [25]: def x():  
         print("yasir")  
  
         say_hello = x  
  
         say_hello()  
  
         x()
```

yasir
yasir

```
In [26]: def add(a: int,b: int) -> int:  
         return a + b  
         result = add(3,4)  
         print(result)
```

7

```
In [27]: #A decorator is a function that takes another function as an argument, ad  
         #It's used to "wrap" another function with extra behavior – like logging,  
  
         def decorator(func):  
             def x():  
                 print("Hii")  
                 func()  
                 print("yasir")  
             return x  
  
         @decorator  
         def say_hi():  
             print("Arafat")  
  
         say_hi()
```

Hii
Arafat
yasir

```
In [28]: user = {
    "username": "Yasir",
    "is_logged_in": False
}

def login_required(func):
    def wrapper():
        if user["is_logged_in"]:
            return func()
        else:
            print("Access Denied! Please login first.")
    return wrapper

@login_required
def view_dashboard():
    print(f"Welcome {user['username']}! This is your dashboard.")

def login(username):
    if username == user["username"]:
        user["is_logged_in"] = True
        print(f"{username} logged in successfully!")
    else:
        print("Invalid username!")

view_dashboard()

login("Yasir")

view_dashboard()
```

Access Denied! Please login first.
Yasir logged in successfully!
Welcome Yasir! This is your dashboard.

In []: